Assessment Instructions

Time Limits: 3 hours

- This assessment is individual and not proctored. By submitting, you confirm the work is your own. You may use standard references and documentation. Please disclose any AI tools used and describe how.
- Shortlisted candidates will be asked to walk through their code and answer questions about their approach in a technical call.

You are given an input file named: “Generator Offer Curve Data for Assessment.csv”

The file contains hourly offer curve data for a generator in ERCOT energy market for one year. Each row represents one hour. An offer curve is a monotonically increasing set of price-quantity pairs that indicates how much energy (in MW) a given generation resource is willing to sell at or above each of those price levels, up to the resource’s high limit.

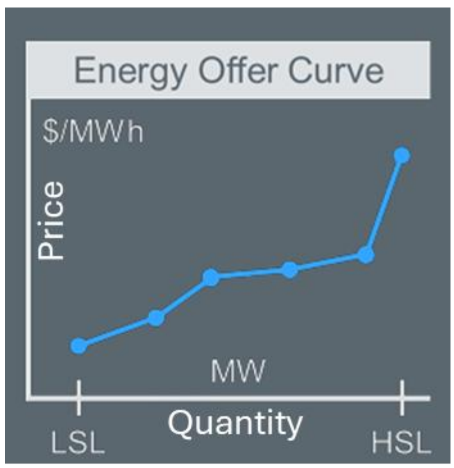

Figure: Offer Curve for one Hours

The offer curve is represented by MW/price breakpoints:

Quantity-MW1, Price1

Quantity-MW2, Price2

…

Quantity-MW13, Price13

Your task is to write code that reads the input file, cleans the data, and creates an interactive plot showing the generator's offer curves by **hour of day**.

Expected Output

Create a single interactive plot with 24 subplots (for 24 hours) arranged in a 6-by-4 grid.

Each subplot should represent one hour of the day:

Hour 00, Hour 01, ..., Hour 23

For each hour subplot:

- Plot **all** historical daily offer curves for that hour in a light blue or transparent style.

- Overlay a forecasted offer curve for the target forecast date **(05/29/2026)** as a solid line.

- Show the forecasted MW/price breakpoints as dots on the line.

- The x-axis should be MW.

- The y-axis should be offer price.

The plot may be delivered in one of the following formats:

- Streamlit (preferred)
- HTML with JavaScript
- Jupyter Notebook charts

Forecast Requirement

Use the historical data to create a forecast offer curve for each hour of the target day.

A simple acceptable forecasting method is:

1. Use the same calendar date from the prior year if available.

2. If that exact date is not available, use the median curve for the same hour, month, and weekday.

3. If that is not available, use the median curve for the same hour across the full dataset.

4. Use any other method that would provide a good forecast.

Data Cleaning Requirements

The input file contains data quality issues. Your code should detect and clean these issues before plotting.

At minimum, your cleaning process should handle the following:

1. Invalid timestamps

Some rows may contain invalid timestamp values.

You should:

- Convert the `Timestamp` column to a proper datetime field.

- Remove rows where the timestamp cannot be parsed.

- Sort the final dataset by timestamp.

2. Duplicate timestamps

There may be duplicate hourly records.

You should:

- Detect duplicate `Timestamp` values.

- Keep one valid record per timestamp.

- Remove duplicate timestamp rows before plotting.

3. Duplicate rows

The file may contain repeated rows.

You should:

- Detect exact duplicate rows.

- Remove repeated rows.

4. Numeric conversion issues

Some numeric fields may contain text or blank values.

This may occur in columns such as:

Quantity-MW1 … Quantity-MW13

Price1 … Price13

High Limit

Low Limit

You should:

- Convert MW and price columns to numeric values.

- Treat non-numeric values as missing or invalid.

- Exclude invalid MW/price breakpoints from the curve.

5. Missing curve points

Some offer curve breakpoints may be partially or fully missing.

You should:

- Drop individual MW/price breakpoint pairs where either MW or price is missing.

- Do not necessarily drop the entire hourly row if enough valid curve points remain.

6. Extreme price outliers

The file may contain unrealistic offer prices, including very large positive or negative values.

You should:

- Identify prices that are clearly outside a reasonable range.

- Remove or correct those individual price points.

- Document the thresholds or logic you used.

7. Impossible MW values

The file may contain MW values that are not physically meaningful.

You should:

- Remove negative MW values.

- Remove extremely large MW values that are clearly not plausible for this unit.

- Document the threshold or logic you used.

8. Non-monotonic MW breakpoints

Offer curve MW breakpoints should generally increase as the curve progresses.

The file may contain rows where adjacent MW breakpoints are out of order.

You should:

- Detect non-monotonic MW breakpoints.

- Clean them by sorting valid breakpoint pairs by MW, or by removing the offending point.

- Ensure the plotted curve does not loop backward on the x-axis.

9. High Limit / Low Limit issues

Some rows may have inconsistent limit values.

You should:

- Convert `High Limit` and `Low Limit` to numeric values.

- Check whether `Low Limit` is greater than `High Limit`.

- Correct obviously swapped limits or remove invalid limit records.

- Document your approach.

Suggested Implementation Steps

1. Load the CSV file.

2. Rename or standardize columns if helpful.

3. Parse `Timestamp`.

4. Remove invalid and duplicate timestamp records.

5. Convert all MW, price, and limit columns to numeric values.

6. The offer curve for each hour is stored across several pairs of columns, such as Quantity-MW1 / Price1, Quantity-MW2 / Price2, and so on. Convert these pairs into a table where each MW/price pair becomes one row. This will make it easier to clean, analyze, and plot the offer curves.

7. Remove invalid curve points.

8. Sort curve points by timestamp, hour, and MW.

9. Build historical offer curves for each hour of day.

10. Create forecast curves for the target date.

11. Generate the 6-by-4 plot.

Deliverables

Use the google form below to submit the following information. Also submit the same information via email to srv@newgridconsulting.com.

https://forms.gle/2f9cYXsdmSc7bYq97

- Your code along with the requirements.txt file via a public link to the project's GitHub repository **or** a drive location where the code is stored. Provide a readme.md file to setup the project and generate the plots.

- Also provide a URL for the published website for the forecast (preferrable) **or** a snapshot/video of the generated plot.

- A short explanation of your data cleaning approach.

- A short explanation of your forecast method.

- Any assumptions you made.

Evaluation Criteria

Your submission will be evaluated on:

- Correctly reading and reshaping the input data.
- Robust handling of dirty data and outliers.
- Clear and defensible data cleaning choices.
- Producing the requested 24-panel plot.
- Clearly showing the forecast curve and its MW/price points.
- Code readability and organization.
- Clear explanation of assumptions and methodology.

In case you have any questions, please contact: srimanvgn@gmail.com